Yunsung Chung

Computer Science Ph.D. Researcher | Applied Machine Learning
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EXPERIENCE

Samsung Research America – Research Intern May 2025 – Aug 2025
Mountain View, CA | Supervisor: Dr. Sharanya Desai

- Designed a **biomarker-specific temporal-decay loss** to learn from dense smartwatch pulse-wave (PPG) signals and sparse clinical labels while accounting for label staleness.
- Built a PPG signal-quality index (SQI) algorithm applied to **80K+ hours of PPG from 50K participants**, enabling task-specific filtering and consistent data-quality checks across experiments.
- Contributed biomarker-specific decay-loss methods to a **filed patent application (patent pending)**.

Tulane University, TRIAD Lab – Research Assistant Aug 2022 – Present
New Orleans, LA | Supervisor: Dr. Jihun Hamm

- Developed **CRAFT**, combining clinical rewards with diffusion fine-tuning across four medical-imaging datasets; earned a **110% higher physician top-choice rate than SOTA** in a blinded evaluation.
- Developed an intervention-aware world model of longitudinal patient-state evolution after ablation (**MWM 2026**, oral); improved **recurrence-prediction AUROC by 15.8% over SOTA**.
- Developed **SOFA**, a deep-learning framework for simulating and optimizing atrial fibrillation ablation strategies (**MICCAI 2025**).
- Created and released **EHRDyn**, a unified benchmark spanning **five ICU care domains**; designed layered evaluation of patient-state prediction, treatment-policy planning, and off-policy value estimation across real EHRs and complementary simulators (**KDD 2027**).
- Built **EHRGym**, a clinical simulation benchmark that identifies reinforcement-learning failure modes under missing measurements, irregular sampling, and repeated clinical records.
- Contributed to a **PPG-based cardiac rhythm monitoring model**, using knowledge distillation to transfer ECG-derived rhythm information to **PPG-only inference**.
- Developed a **three-stage evaluation framework** and led empirical benchmarking of synthetic images as substitutes for real training data, measuring privacy–utility trade-offs when sharing images or trained classifiers (**USENIX Security 2025**).

Airport Railroad Express – Hardware Engineer Nov 2017 – May 2019
Incheon, South Korea

- Tested and debugged automated fare-collection and passenger-service hardware for a high-reliability transportation system.

EDUCATION

Tulane University Aug 2022 – Present
Ph.D. in Computer Science (in progress) New Orleans, LA

The Australian National University Jul 2019 – Jul 2021
M.S. in Computer Science Canberra, Australia

Dongguk University Mar 2012 – Feb 2018
B.S. in Electronics, Information & Communication Engineering Seoul, South Korea

TECHNICAL SKILLS

Tools: Python, PyTorch, Hugging Face, NumPy, SciPy, pandas, Scikit-learn, SQL, Git, AWS, Weights & Biases.
Methods: Time-series models, Transformers, multimodal learning, diffusion models, LoRA/PEFT, DPO-based diffusion adaptation, contrastive learning, offline RL, off-policy evaluation, uncertainty estimation.
Data: Longitudinal EHR, ECG/PPG, medical imaging, sparse clinical labels.

RESEARCH SOFTWARE

- **ResearchWiki / ResearchOMX:** Built an LLM-assisted research workflow with **41 automated evidence and consistency checks**, claim-to-source links, and five review roles. Used Git and a Markdown wiki to maintain reproducible experiment records and auditable research decisions.
- **EHRDyn public benchmark:** Released versioned task definitions, reusable submission and scoring interfaces, validation tests, and reproducible reports so methods can be compared under a consistent evaluation protocol. Added explicit unsupported-result handling and fixed data roles to make evaluation failures visible and comparisons reproducible.

SELECTED PUBLICATIONS

Peer-Reviewed Publications

[1] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *EHRDyn: A Benchmark for Offline Reinforcement Learning from EHRs with Complementary Simulators.* **KDD** 2027.

[2] Yingshuo Liu, Chanho Lim, Ghassan Bidaoui, Christian Massad, **Yunsung Chung**, et al. *Reclassifying Single Lead Sinus Rhythm ECGs for Improved AF Recurrence Detection With Deep Learning.* **Circulation: Arrhythmia and Electrophysiology**, e014659, 2026. doi:10.1161/CIRCEP.125.014659.

[3] **Yunsung Chung**, Yingshuo Liu, Abboud F. Hassan, Han Feng, Mary M. Maleckar, Nassir Marrouche, and Jihun Hamm. *Intervention-Aware Clinical World Model for Post–Op Outcome Forecasting in Cardiology.* **MICCAI Workshop on Medical World Models (MWM)**, 2026. **Oral presentation**.

[4] **Yunsung Chung**, Keum San Chun, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanho Lim, Viswam Nathan, Nassir Marrouche, and Sharanya Arcot Desai. *Weighted Temporal Decay Loss for Learning Wearable PPG Data with Sparse Clinical Labels.* **ICASSP**, 2026.

[5] **Yunsung Chung**, Chanho Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, and Jihun Hamm. *SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation.* **MICCAI**, 2025.

[6] **Yunsung Chung**, Yunbei Zhang, Nassir Marrouche, and Jihun Hamm. *SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation.* **USENIX Security**, 2025.

[7] **Yunsung Chung**, Janet Wang, and Jihun Hamm. *Bridging the Gap: Synthetic Data Augmentation through Inversion and Distribution Matching for Few-Shot Learning.* **AI-ML Systems**, 2024.

[8] Janet Wang, **Yunsung Chung**, Zhengming Ding, and Jihun Hamm. *From Majority to Minority: A Diffusion-Based Augmentation for Underrepresented Groups in Skin Lesion Analysis.* **MICCAI Workshop**, 2024.

[9] **Yunsung Chung**, Chanho Lim, Chao Huang, Nassir Marrouche, and Jihun Hamm. *FBA-Net: Foreground and Background Aware Contrastive Learning for Semi-Supervised Atrium Segmentation.* **MICCAI Workshop**, 2023.

Manuscripts Under Review

[10] **Yunsung Chung**, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, and Jihun Hamm. *CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis.* Under review at **ICLR**, 2027.

[11] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *EHRGym: A Diagnostic Benchmark for Offline Reinforcement Learning.* Under review at **ICLR**, 2027.

[12] Yingshuo Liu, Alex El Darzi, **Yunsung Chung**, et al. *Deep Learning-Based Translation from Wearable ECG to PPG for PPG-Only Cardiac Rhythm Classification.* Under review at **npj Digital Medicine**, 2026.

AWARDS AND SERVICE

Kaggle: Competitions Expert with four bronze medals across 10 completed competitions (seven solo, three team): Image Matching 2022 (96/642), Sartorius (116/1,505), Happywhale (127/1,588), and PetFinder Pawpularity (238/3,537).

Reviewer: ICASSP 2027; NeurIPS 2026; ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.

Robotics: FIRA RoboWorld Cup 2016, 1st Place; International Robot Contest 2016, 2nd Place.

Outreach and Scholarships: NeurIPS 2023 College Outreach Volunteer; First-Class Academic Excellence Scholarship, Dongguk University, 2012 and 2015–2017.